

Per-position mutation frequency

random

dpo

P(non-WT residue)

1 H 2 M 3 S 4 M 5 Q 6 Q 7 V 8 V 9 S 10 A 11 G 12 W 13 E 14 R 15 A 16 D 17 L 18 V 19 G 20 D 21 A 22 F 23 D 24 V

CDR-H3 position (1-indexed; WT residue beneath)